Joyent[34] has been championing this approach, with several[35] compelling articles[36] on the subject. It also provides an open source tool called ContainerPilot[37] to help with the coordination of life-cycle events. Alternatively, you can roll your own solution. I suspect we'll see more in this space over time.

Database Replication and High Availability

To replicate your database, you typically need to rely on the built-in capabilities of your database (rather than a super naive approach of trying to use a shared filesystem).

Postgres offers many different options for clustering.[38] Rather than reinventing the wheel, however, you can leverage the work that others have done in this area—for example:

- Patroni[39]
- Barman[40]
- Crunchy[41]

You'll find similar work has been done for clustering and replicating other databases too.

34. https://www.joyent.com
35. https://www.joyent.com/blog/app-centric-micro-orchestration
36. https://www.joyent.com/blog/persistent-storage-patterns
37. https://www.joyent.com/containerpilot
38. https://wiki.postgresql.org/wiki/Replication,_Clustering,_and_Connection_Pooling
39. https://github.com/zalando/patroni
40. https://hub.docker.com/r/postdock/barman/
41. http://info.crunchydata.com/blog/an-easy-recipe-for-creating-a-postgresql-cluster-with-docker-swarm